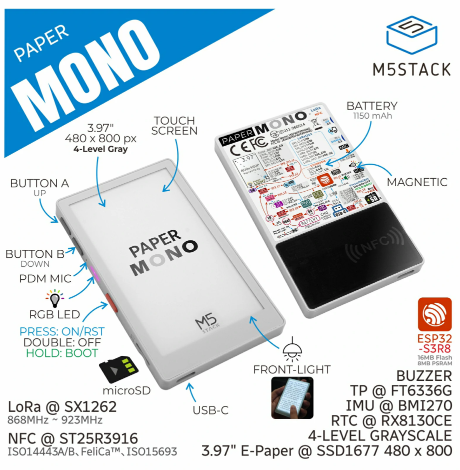
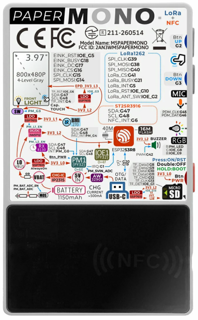
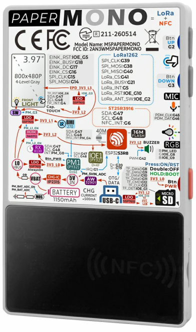
PaperMono

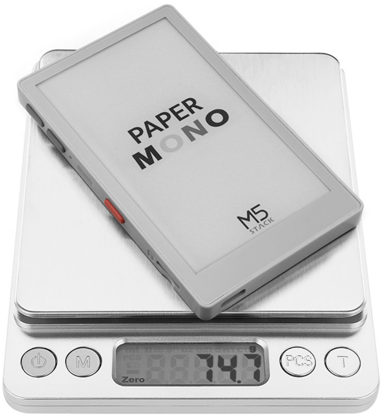
SKU:C153


PAPER
MONO
M5STACK
3.97"
480 x 800 px
4-Level Gray
TOUCH SCREEN
BATTERY
1150 mAh
MAGNETIC
BUTTON A
UP
BUTTON B
DOWN
PDM MIC
RGB LED
PRESS: ON/RST
DOUBLE: OFF
HOLD: BOOT
microSD
USB-C
FRONT-LIGHT
ESP32
-S3R8
16MB Flash
8MB PSRAM
BUZZER
TP @ FT6336G
IMU @ BMI270
RTC @ RX8130CE
4-LEVEL GRAYSCALE
3.97" E-Paper @ SSD1677 480 x 800
LoRa @ SX1262
868MHz ~ 923MHz
NFC @ ST25R3916
ISO14443A/B、FeliCa™、ISO15693



PAPER
MONO
M5 STACK

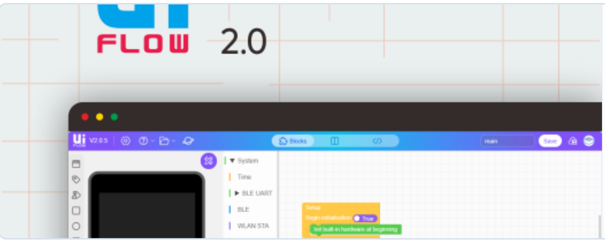




Description

PaperMono is an e-paper development board built around low-power display, near-field identification, and long-range communication capabilities. It features a touch-enabled 3.97-inch 4-level grayscale monochrome e-paper display, driven by SSD1677 with a resolution of 480 x 800, and integrates an FT6336G touch controller and a frontlight system for a more comfortable reading and interaction experience in low-light environments. The core adopts an ESP32-S3R8, paired with 16MB Flash, 8MB PSRAM, and 2.4 GHz Wi-Fi, along with an onboard buzzer, PDM microphone (LMD4737T261-AC02), RGB LED, BMI270 6-axis IMU, M5IOE1 IO expander, microSD card slot, and RX8130CE real-time clock. The NFC section is based on the ST25R3916, supporting ISO14443A, ISO14443B, FeliCa™, and ISO15693; LoRa communication adopts the SX1262 (Stamp LoRa-1262) solution with a built-in FPC antenna, supporting the 868MHz ~ 923MHz band. Combined with the M5PM1 multi-level power management system and a 1150mAh battery, it meets the requirements of low-power IoT and embedded applications such as e-readers, e-paper signage, access control terminals, identity authentication, and intelligent transportation.

Tutorial

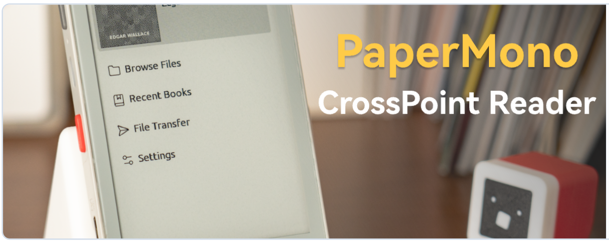


UiFlow2

This tutorial will introduce you to controlling the PaperMono device through the UiFlow2 graphical programming platform.

Arduino IDE

This tutorial introduces how to program and control the PaperMono device using Arduino IDE.



CrossPoint E-Reader

Click here to download the PaperMono CrossPoint E-Reader firmware.

Note

E-Paper Usage Precautions

1. Avoid direct sunlight/prolonged sun exposure during use; high temperatures or strong UV can damage the panel.
2. After approximately 10 partial fast refreshes in software, it is recommended to perform one full-screen refresh to clear the screen and avoid severe accumulated ghosting.
3. The screen has built-in OTP waveforms that can be called directly. When using custom external waveforms, pay attention to DC balance; otherwise, irreversible damage may be caused to the panel.
4. Avoid uninterrupted continuous partial fast refreshes, to prevent long-term DC imbalance from causing irreversible damage to the panel.

Touch Active Area

PaperMono Touch Active Area

The touch IC firmware internally shrinks the touch boundary. The effective touch coordinate range is limited to: **X-axis: 5 ~ 475** (total width 480px), **Y-axis: 5 ~ 795** (total height 800px).

Features

- ESP32-S3R8 Core Controller
  - 16MB Flash
  - 8MB PSRAM
  - 2.4 GHz Wi-Fi
- 3.97-inch 4-level grayscale monochrome e-paper display with touch
  - SSD1677
  - 480x800 resolution
  - 4-level grayscale display
  - FT6336G touch
- Integrated frontlight system
- Audio interaction
  - PDM microphone (LMD4737T261-AC02)
  - Built-in buzzer
- Onboard peripherals
  - BMI270 6-axis IMU
  - M5IOE1 IO expander
  - microSD card slot
  - RX8130CE RTC

    - RGB LED
- NFC function
    - ST25R3916
    - Supports ISO14443A / ISO14443B / FeliCa™ / ISO15693
- LoRa communication
    - SX1262 (Stamp LoRa-1262)
    - 868MHz ~ 923MHz
    - Built-in FPC antenna
- Built-in 1150mAh battery

Includes

- 1 x PaperMono

Applications

- E-Reader
- E-Paper Signage
- Access Control Terminal
- Identity Authentication Device
- Intelligent Transportation Terminal

Specifications

Specification	Parameter
SoC	ESP32-S3R8 @ Xtensa® 32-bit LX7 dual-core processor, 240MHz
Flash	16MB
PSRAM	8MB Octal
Wi-Fi	2.4 GHz Wi-Fi
Screen	3.97" E-Paper (Mono) SSD1677 @ 480x800, Touch FT6336G
Screen Grayscale	4-level grayscale display
Frontlight	Integrated e-paper frontlight
Input Power	USB Type-C DC 5V
Battery Capacity	1150mAh
Microphone	PDM microphone LMD4737T261-AC02
Buzzer	Built-in buzzer
IMU	BMI270
Power Management	M5PM1
IO Expander	M5IOE1
Expansion Storage	microSD
RTC	RX8130CE
RGB LED	Built-in RGB LED
NFC	ST25R3916 (ISO14443A/B, FeliCa™, ISO15693)
LoRa	SX1262 (Stamp LoRa-1262), 868MHz ~ 923MHz, built-in FPC antenna
User Buttons	2x user buttons + 1x power button (ON / OFF / RESET / BOOT)
Product Size	62.0 x 101.0 x 8.0mm
Product Weight	74.7g
Package Size	113.2 x 69.6 x 21.0mm
Gross Weight	91.3g

Learn

Power On/Off

- Power on / Reset: press the power button once briefly
- Power off: press the power button twice in succession

C153_PaperMono_poweron_EN.mp4

Download Mode

Press and hold the power button (about 2 seconds) until the red LED blinks, then release. The device enters download mode and is ready for flashing.

M5GFX LUT Refresh Speed

The following data is the laboratory test result of PaperMono under different M5GFX refresh modes. The actual refresh time may vary depending on the display content and operating environment, and is provided for reference only.

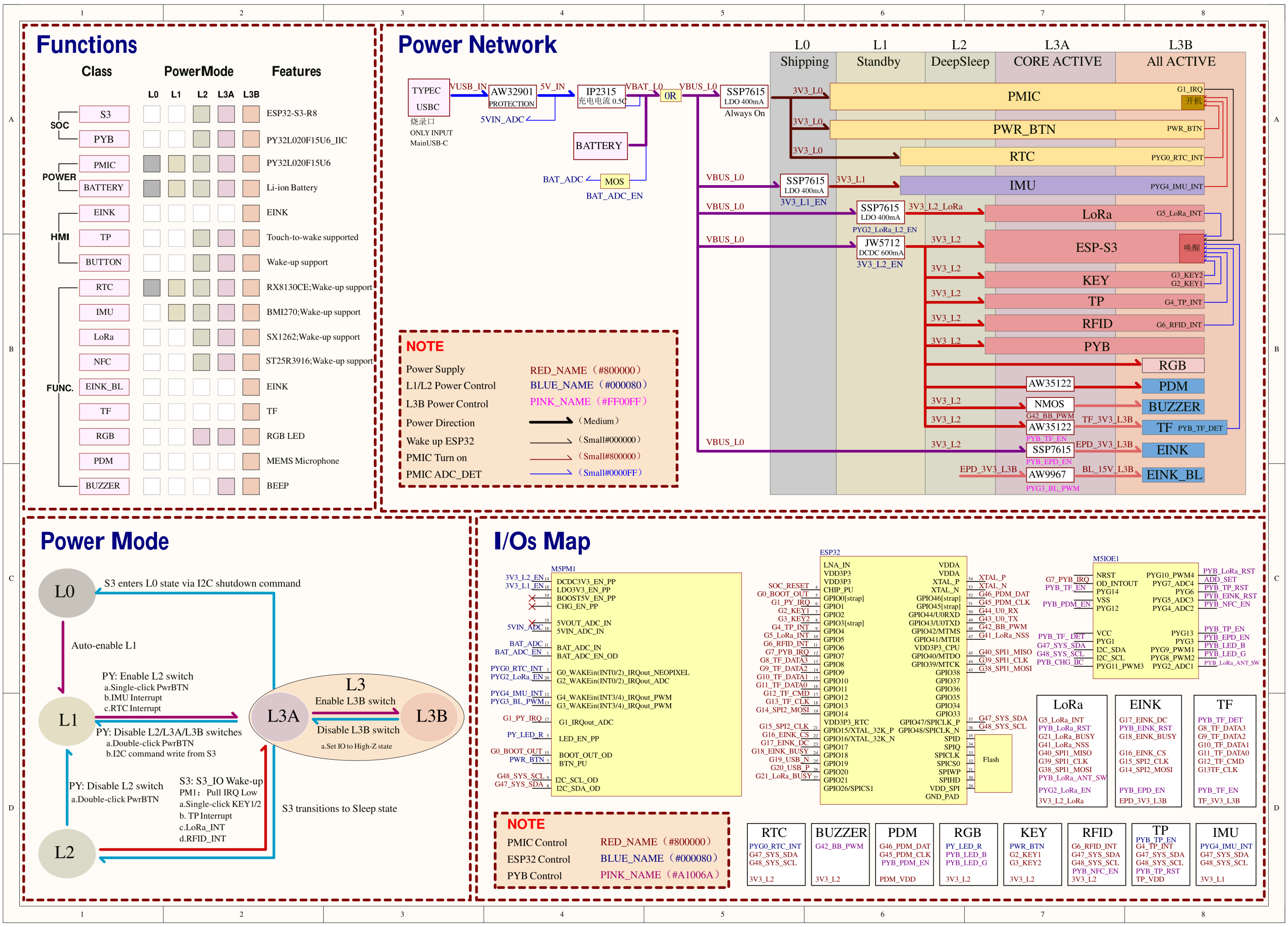
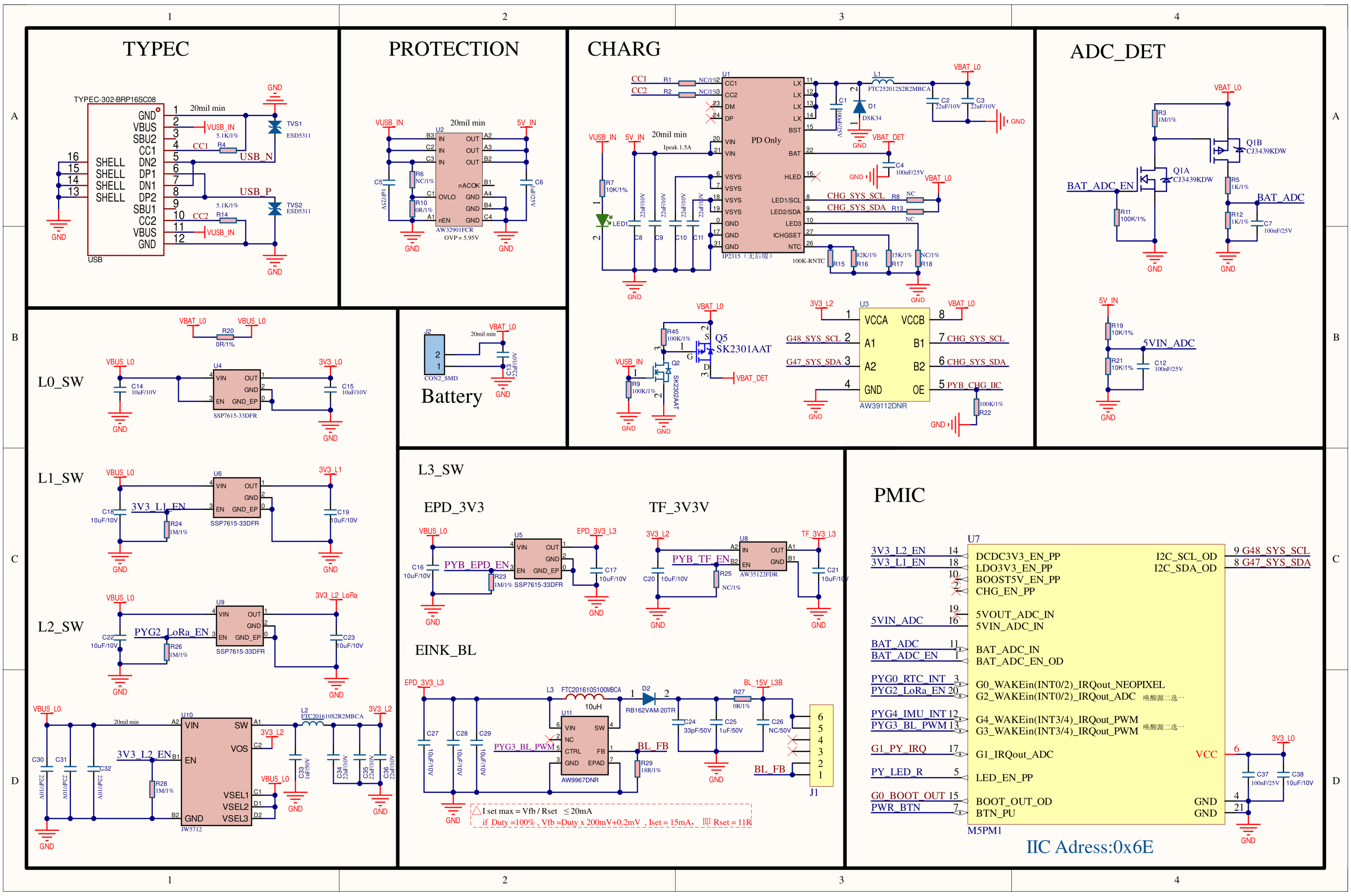
Refresh Mode	Time per Refresh
`epd_quality`	4.71 s
`epd_text`	0.45 s
`epd_fast`	0.34 s
`epd_fastest`	0.07 s

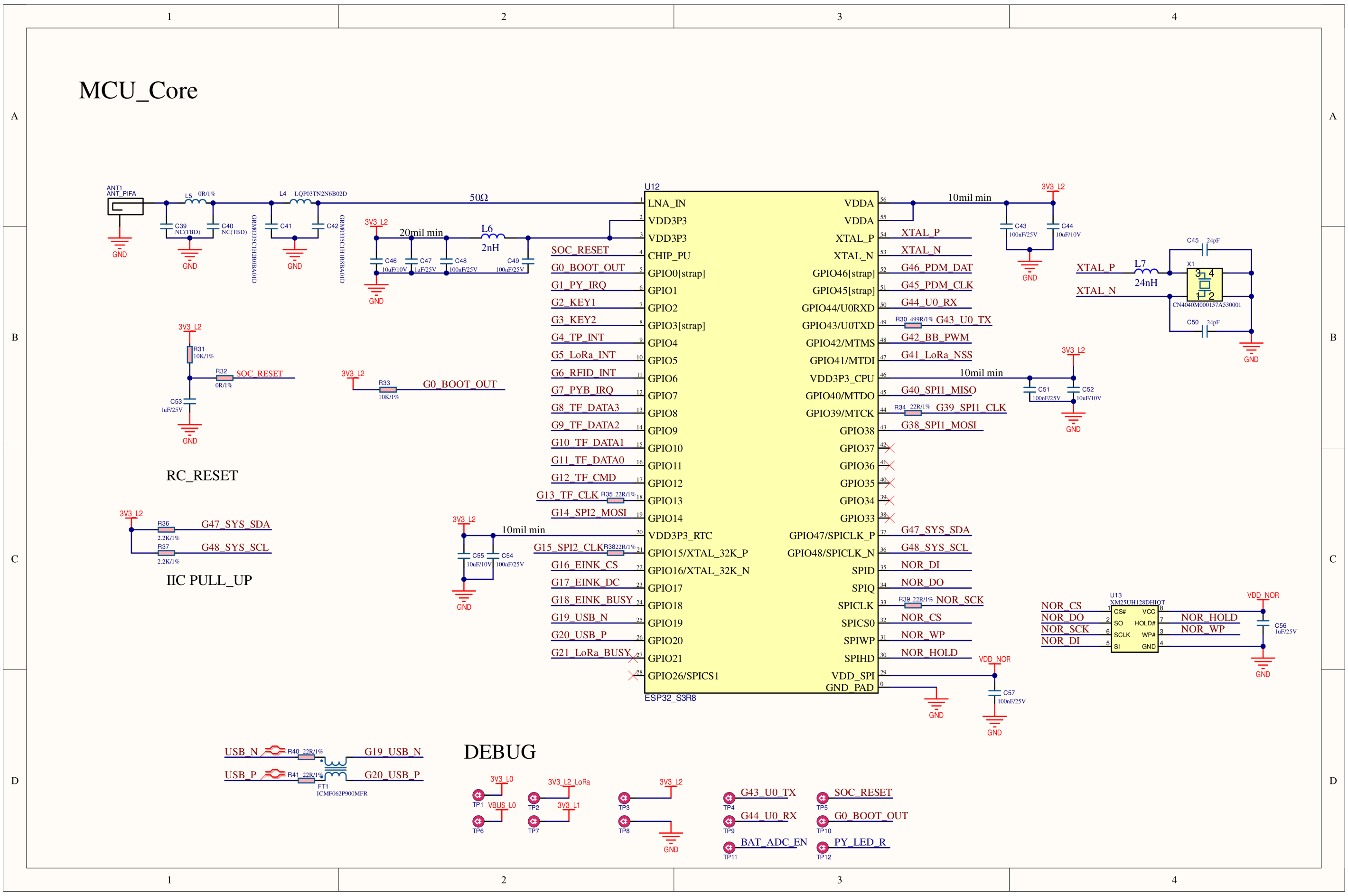
Refresh Mode Demonstration:

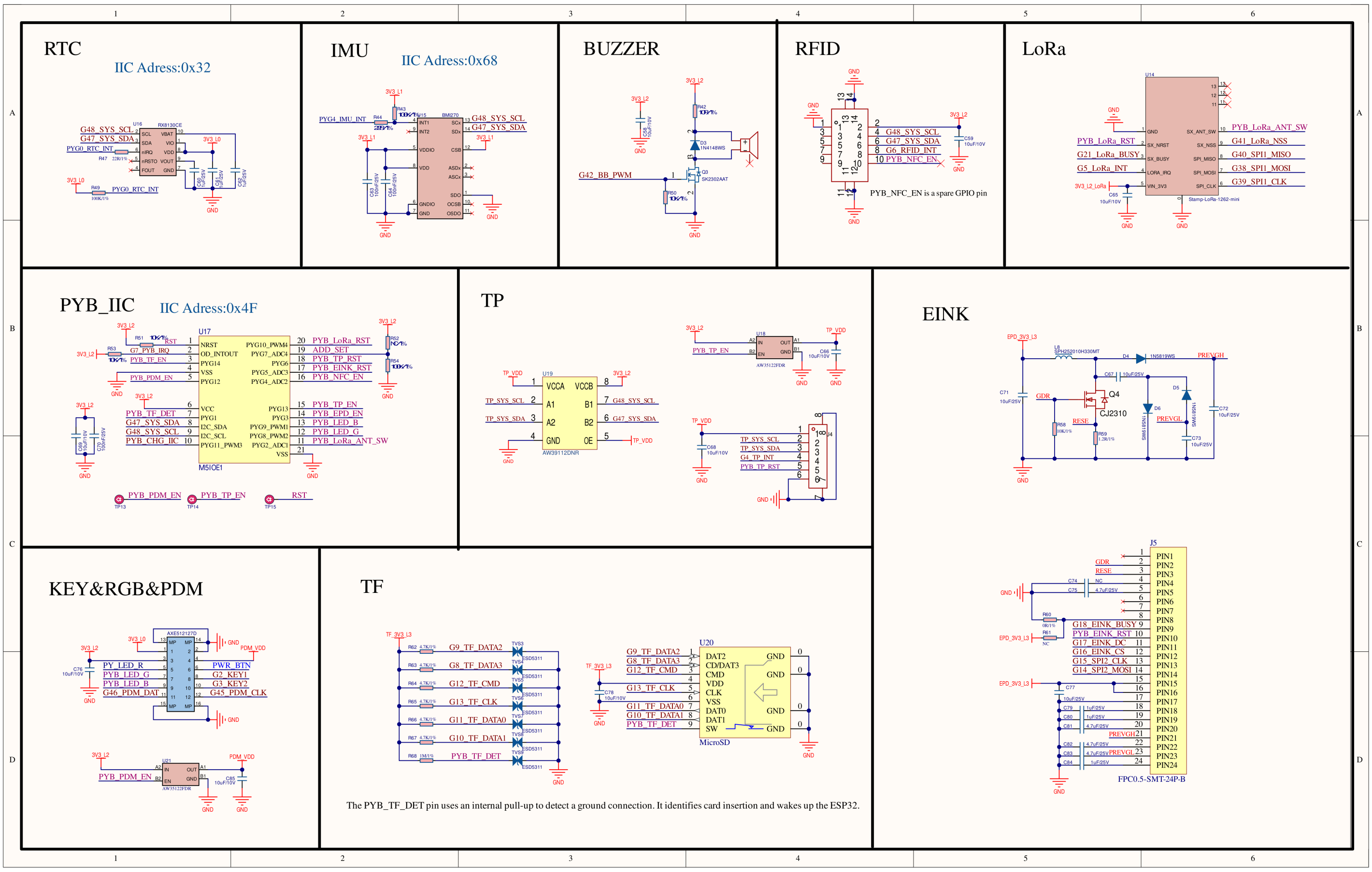
C153_C153-Lite_reflash_mode_demo.mp4

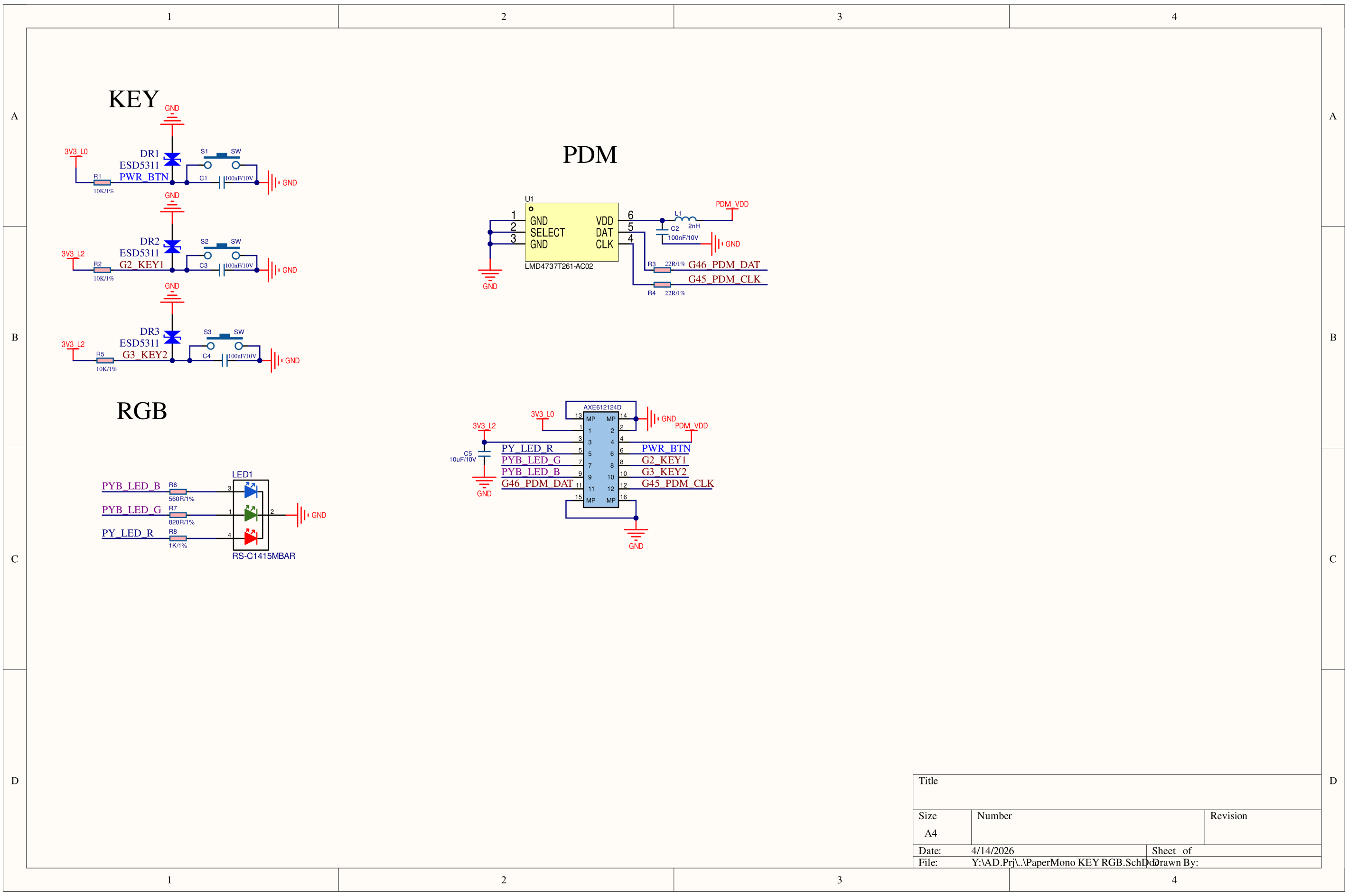
Schematics

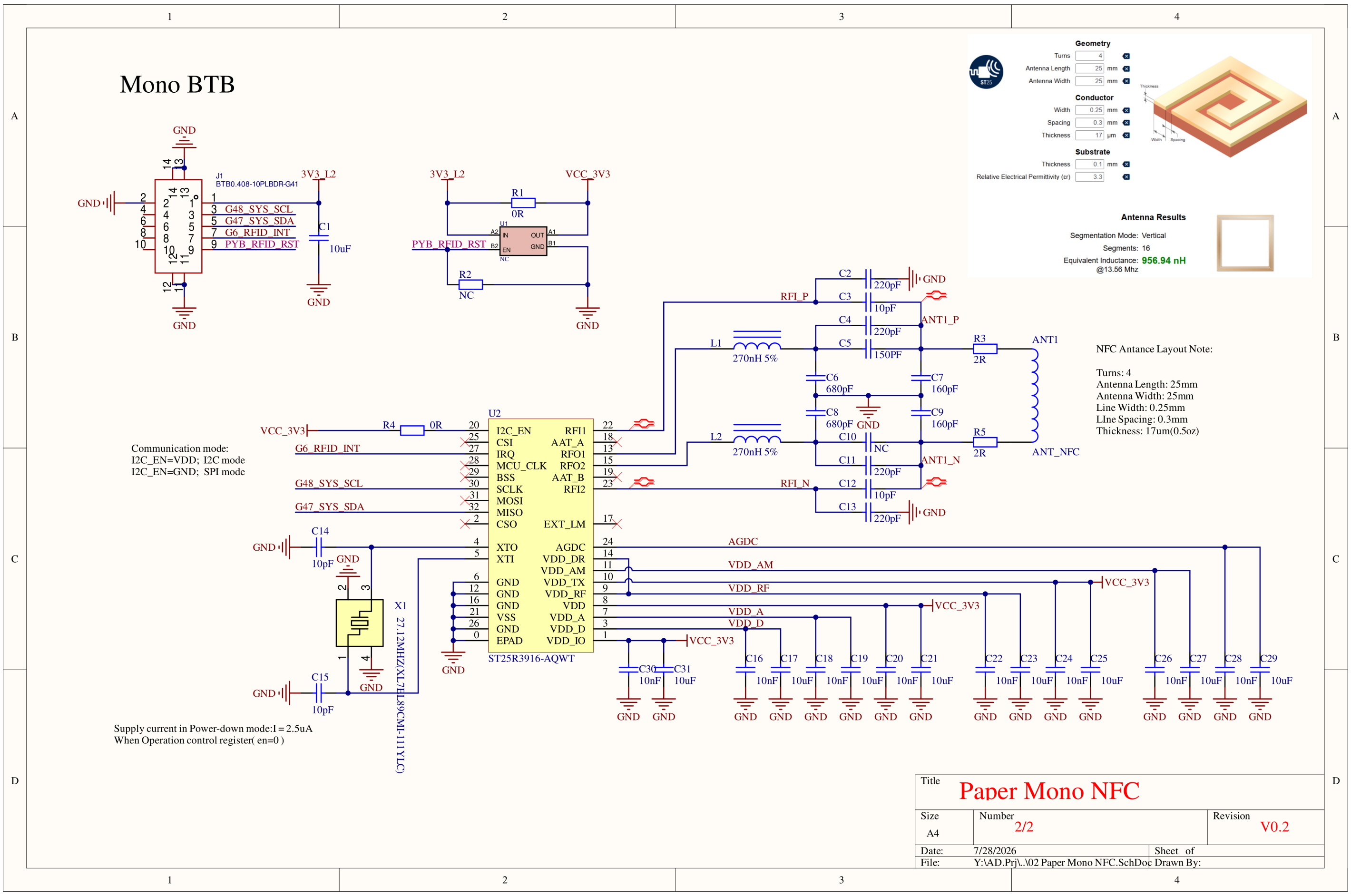
- PaperMono Schematics PDF


Functions
Class
PowerMode
Features
L0 L1 L2 L3A L3B
SOC
S3
ESP32-S3-R8
PYB
PY32L020F15U6_IIC
POWER
PMIC
PY32L020F15U6
BATTERY
Li-ion Battery
HMI
EINK
EINK
TP
Touch-to-wake supported
BUTTON
Wake-up support
FUNC.
RTC
RX8130CE;Wake-up support
IMU
BMI270;Wake-up support
LoRa
SX1262;Wake-up support
NFC
ST25R3916;Wake-up support
EINK_BL
EINK
TF
TF
RGB
RGB LED
PDM
MEMS Microphone
BUZZER
BEEP
Power Network
L0 Shipping
L1 Standby
L2 DeepSleep
L3A CORE ACTIVE
L3B All ACTIVE
TYPEC USBC
烧录口
ONLY INPUT
MainUSB-C
AW32901 PROTECTION
IP2315 充电电流 0.5A
BATTERY
MOS
SSP7615 LDO 400mA Always On
PMIC
PWR_BTN
RTC
IMU
LoRa
ESP-S3
KEY
TP
RFID
PYB
RGB
PDM
BUZZER
TF
EINK
EINK_BL
NOTE
Power Supply — RED_NAME（#800000）
L1/L2 Power Control — BLUE_NAME（#000080）
L3B Power Control — PINK_NAME（#FF00FF）
Power Direction — （Medium）
Wake up ESP32 — （Small#000000）
PMIC Turn on — （Small#800000）
PMIC ADC_DET — （Small#0000FF）
Power Mode
L0
S3 enters L0 state via I2C shutdown command
Auto-enable L1
L1
PY: Enable L2 switch
a.Single-click PwrBTN
b.IMU Interrupt
c.RTC Interrupt
PY: Disable L2/L3A/L3B switches
a.Double-click PwrBTN
b.I2C command write from S3
L3
L3A
L3B
Enable L3B switch
Disable L3B switch
a.Set IO to High-Z state
PY: Disable L2 switch
a.Double-click PwrBTN
S3: S3_IO Wake-up
PM1: Pull IRQ Low
a.Single-click KEY1/2
b. TP Interrupt
c.LoRa_INT
d.RFID_INT
S3 transitions to Sleep state
L2
I/Os Map
M5PM1
ESP32
M5IOE1
LoRa
EINK
TF
NOTE
PMIC Control — RED_NAME（#800000）
ESP32 Control — BLUE_NAME（#000080）
PYB Control — PINK_NAME（#A1006A）
RTC
BUZZER
PDM
RGB
KEY
RFID
TP
IMU

TYPEC
TYPEC-302-BRP16SC08
PROTECTION
AW32901FCR
OVP = 5.95V
CHARG
IP2315
ADC_DET
BAT_ADC_EN
BAT_ADC
L0_SW
SSP7615-33DFR
L1_SW
3V3_L1_EN
L2_SW
PYG2_LoRa_EN
3V3_L2_EN
JW5712
Battery
L3_SW
EPD_3V3
PYB_EPD_EN
TF_3V3V
PYB_TF_EN
AW35122FDR
EINK_BL
AW9967DNR
PYG3_BL_PWM
I set max = Vfb / Rset ≤ 20mA
if Duty =100% , Vfb =Duty x 200mV+0.2mV , Iset = 15mA, 即 Rset = 11R
AW39112DNR
PMIC
M5PM1
IIC Adress:0x6E


MCU_Core
U12
ESP32_S3R8
LNA_IN
VDD3P3
CHIP_PU
GPIO0[strap]
VDDA
XTAL_P
XTAL_N
GPIO46[strap]
GPIO45[strap]
GPIO44/U0RXD
GPIO43/U0TXD
VDD3P3_CPU
VDD3P3_RTC
VDD_SPI
GND_PAD
50Ω
20mil min
10mil min
2nH
24nH
SOC_RESET
G0_BOOT_OUT
G1_PY_IRQ
G2_KEY1
G3_KEY2
G4_TP_INT
G5_LoRa_INT
G6_RFID_INT
G7_PYB_IRQ
G8_TF_DATA3
G9_TF_DATA2
G10_TF_DATA1
G11_TF_DATA0
G12_TF_CMD
G13_TF_CLK
G14_SPI2_MOSI
G15_SPI2_CLK
G16_EINK_CS
G17_EINK_DC
G18_EINK_BUSY
G19_USB_N
G20_USB_P
G21_LoRa_BUSY
G46_PDM_DAT
G45_PDM_CLK
G44_U0_RX
G43_U0_TX
G42_BB_PWM
G41_LoRa_NSS
G40_SPI1_MISO
G39_SPI1_CLK
G38_SPI1_MOSI
G47_SYS_SDA
G48_SYS_SCL
NOR_DI
NOR_DO
NOR_SCK
NOR_CS
NOR_WP
NOR_HOLD
VDD_NOR
U13
3V3_L2
GND
RC_RESET
IIC PULL_UP
DEBUG
USB_N
USB_P
FT1
TP1
TP2
TP3
TP4
TP5
TP6
TP7
TP8
TP9
TP10
TP11
TP12
3V3_L0
VBUS_L0
3V3_L2_LoRa
3V3_L1
BAT_ADC_EN
PY_LED_R
ANT1
ANT_PIFA
X1


RTC
IIC Adress:0x32
IMU
IIC Adress:0x68
BUZZER
RFID
LoRa
PYB_IIC
IIC Adress:0x4F
TP
EINK
KEY&RGB&PDM
TF
PYB_NFC_EN is a spare GPIO pin
The PYB_TF_DET pin uses an internal pull-up to detect a ground connection. It identifies card insertion and wakes up the ESP32.
U14
U15
U16
U17
U18
U19
U20
U21
Q4
CJ2310
J5
FPC0.5-SMT-24P-B
MicroSD
M5IOE1
AW39112DNR
Stamp-LoRa-1262-mini
PYB_LoRa_ANT_SW
G41_LoRa_NSS
G40_SPI1_MISO
G38_SPI1_MOSI
G39_SPI1_CLK
PYB_LoRa_RST
G21_LoRa_BUSY
G5_LoRa_INT
G42_BB_PWM
G48_SYS_SCL
G47_SYS_SDA
G6_RFID_INT
PYB_NFC_EN
PY_LED_R
PYB_LED_G
PYB_LED_B
G46_PDM_DAT
PWR_BTN
G2_KEY1
G3_KEY2
G45_PDM_CLK
PYB_PDM_EN
PYB_TP_EN
RST
G9_TF_DATA2
G8_TF_DATA3
G12_TF_CMD
G13_TF_CLK
G11_TF_DATA0
G10_TF_DATA1
PYB_TF_DET
G18_EINK_BUSY
PYB_EINK_RST
G17_EINK_DC
G16_EINK_CS
G15_SPI2_CLK
G14_SPI2_MOSI
TP_SYS_SCL
TP_SYS_SDA
G4_TP_INT
PYB_TP_RST
TP_VDD
EPD_3V3_L3
GDR
RESE
PREVGH
PREVGL


KEY
PDM
RGB
DR1
ESD5311
PWR_BTN
DR2
ESD5311
G2_KEY1
DR3
ESD5311
G3_KEY2
3V3_L0
3V3_L2
GND
S1
S2
S3
SW
C1
C3
C4
100nF/10V
R1
R2
R5
10K/1%
U1
GND
SELECT
VDD
DAT
CLK
LMD4737T261-AC02
L1
2nH
C2
100nF/10V
PDM_VDD
R3
R4
22R/1%
G46_PDM_DAT
G45_PDM_CLK
AXE612124D
MP
C5
10uF/10V
PY_LED_R
PYB_LED_G
PYB_LED_B
G46_PDM_DAT
PWR_BTN
G2_KEY1
G3_KEY2
G45_PDM_CLK
PYB_LED_B
PYB_LED_G
PY_LED_R
R6
560R/1%
R7
820R/1%
R8
1K/1%
LED1
RS-C1415MBAR
Title
Size
A4
Number
Revision
Date:
4/14/2026
Sheet of
File:
Y:\AD.Prj\..\PaperMono KEY RGB.SchDoc
Drawn By:


Mono BTB
J1
BTB0.408-10PLBDR-G41
G48_SYS_SCL
G47_SYS_SDA
G6_RFID_INT
PYB_RFID_RST
3V3_L2
C1
10uF
GND
R1
0R
VCC_3V3
U1
IN
OUT
EN
GND
NC
R2
NC
Geometry
Turns
Antenna Length
Antenna Width
Conductor
Width
Spacing
Thickness
Substrate
Relative Electrical Permittivity (εr)
Antenna Results
Segmentation Mode: Vertical
Segments: 16
Equivalent Inductance: 956.94 nH
@13.56 Mhz
NFC Antance Layout Note:
Turns: 4
Antenna Length: 25mm
Antenna Width: 25mm
Line Width: 0.25mm
LIne Spacing: 0.3mm
Thickness: 17um(0.5oz)
Communication mode:
I2C_EN=VDD; I2C mode
I2C_EN=GND; SPI mode
R4
0R
U2
I2C_EN
CSI
IRQ
MCU_CLK
BSS
SCLK
MOSI
MISO
CSO
XTO
XTI
GND
VSS
EPAD
RFI1
AAT_A
RFO1
RFO2
AAT_B
RFI2
EXT_LM
AGDC
VDD_DR
VDD_AM
VDD_TX
VDD_RF
VDD
VDD_A
VDD_D
VDD_IO
ST25R3916-AQWT
RFI_P
RFI_N
L1
L2
270nH 5%
C2 220pF
C3 10pF
C4 220pF
C5 150PF
C6 680pF
C7 160pF
C8 680pF
C9 160pF
C10 NC
C11 220pF
C12 10pF
C13 220pF
ANT1_P
ANT1_N
R3 2R
R5 2R
ANT1
ANT_NFC
AGDC
VDD_AM
VDD_RF
VDD_A
VDD_D
C14 10pF
C15 10pF
X1
27.12MHZ(XL7EL89CMI-111YLC)
C16 C17 C18 C19 C20 C21 C22 C23 C24 C25 C26 C27 C28 C29 C30 C31
10nF
10uF
Supply current in Power-down mode:I = 2.5uA
When Operation control register(en=0)
Title
Paper Mono NFC
Size
A4
Number
2/2
Revision
V0.2
Date:
7/28/2026
Sheet of
File:
Y:\AD.Prj\..\02 Paper Mono NFC.SchDoc
Drawn By:

PinMap

E-Paper

ESP32-S3R8	G14_SPI2_MOSI	G15_SPI2_CLK	G16_EINK_CS	G17_EINK_DC	G18_EINK_BUSY
SSD1677	MOSI	SCLK	CS	DC	BUSY

M5IOE1	PYG5_ADC3	PYG3
SSD1677	RST	EPD_3V3_EN

M5PM1	G3_WAKEin / IRQout / PWM
Frontlight	BL_FB

The e-paper reset is controlled via `PYG5_ADC3` of the `M5IOE1`, and `PYG3` controls the 3.3V power supply of the e-paper; the frontlight brightness is controlled via `PYG3_BL_PWM` of the `M5PM1`.

Touch

ESP32-S3R8	G47_SYS_SDA	G48_SYS_SCL	G4_TP_INT
FT6336G(0x38)	SDA	SCL	INT

M5IOE1	PYG6	PYG13
FT6336G	TP_RST	TP_VDD_EN

microSD

ESP32-S3R8	G12	G13	G11	G10	G9	G8
microSD	CMD	CLK	DAT0	DAT1	DAT2	DAT3

M5IOE1	PYG14	PYG1
microSD	TF_EN	TF_DET

`TF_DET` is pulled up by the microSD power domain. When the card slot switch is closed, the detection pin is pulled low to identify microSD card insertion.

HMI

ESP32-S3R8	G47_SYS_SDA	G48_SYS_SCL	G2_KEY1	G3_KEY2
RTC - RX8130CE(0x32)	SDA	SCL		
IMU - BMI270(0x68)	SDA	SCL		
KEY			KEY1	KEY2

M5PM1	G0_WAKEin	G4_WAKEin	LED_EN_PP
HMI	RTC_INT	IMU_INT	LED_R

`PYG0_RTC_INT` and `PYG4_IMU_INT` are connected to the `M5PM1` and can serve as low-power wake-up sources.

M5IOE1	PYG8_PWM2	PYG9_PWM1
HMI	LED_G	LED_B

The RGB LED indicator on the side of the PaperMono body consists of three color LED dies, with the red LED connected to `LED_EN_PP` of the M5PM1. After the device is powered on, it will light up according to the M5PM1 default behavior. Since `LED_EN_PP` does not support PWM output mode configuration, the adjustable colors of this indicator will be limited.

KEY

ESP32-S3R8	G2	G3
KEY	USER_KEY1 (Button A)	USER_KEY2 (Button B)

Audio

ESP32-S3R8	G46_PDM_DAT	G45_PDM_CLK	G42_BB_PWM
PDM MIC	DAT	CLK	
BUZZER			BB_PWM

M5IOE1	PYG12
PDM	PDM_VDD_EN

M5PM1

ESP32-S3R8	G47_SYS_SDA	G48_SYS_SCL	G1_PY_IRQ	G0_BOOT_OUT
M5PM1(0x6E)	SDA	SCL	IRQ	BOOT_OUT

M5IOE1

ESP32-S3R8	G47_SYS_SDA	G48_SYS_SCL	G7
M5IOE1(0x4F)	SDA	SCL	PYB_IRQ

M5IOE1	PYG11_PWM3
Charger - IP2315(0x75)	PYB_CHG_IIC

> **Note**
>
> The I2C operating mode of the charging chip `IP2315` depends on the I2C pins being pulled up to VBAT voltage. Connecting the device to USB triggers mode detection; if the VBAT voltage is too low, the IP2315 may not initialize into I2C mode properly, which can interfere with other devices on the same I2C bus. M5IOE1 controls the connection between the IP2315 and the system I2C bus via PYG11_PWM3: during device operation, `do not keep the IP2315 mounted on the I2C bus for extended periods; disconnect it promptly after communication to avoid reducing bus communication stability`. If an I2C bus communication anomaly occurs, you can reset the device by briefly pressing the power button to restore normal bus operation.

RFID

ESP32-S3R8	G47_SYS_SDA	G48_SYS_SCL	G6_RFID_INT
ST25R3916(0x50)	SDA	SCL	IRQ

LoRa

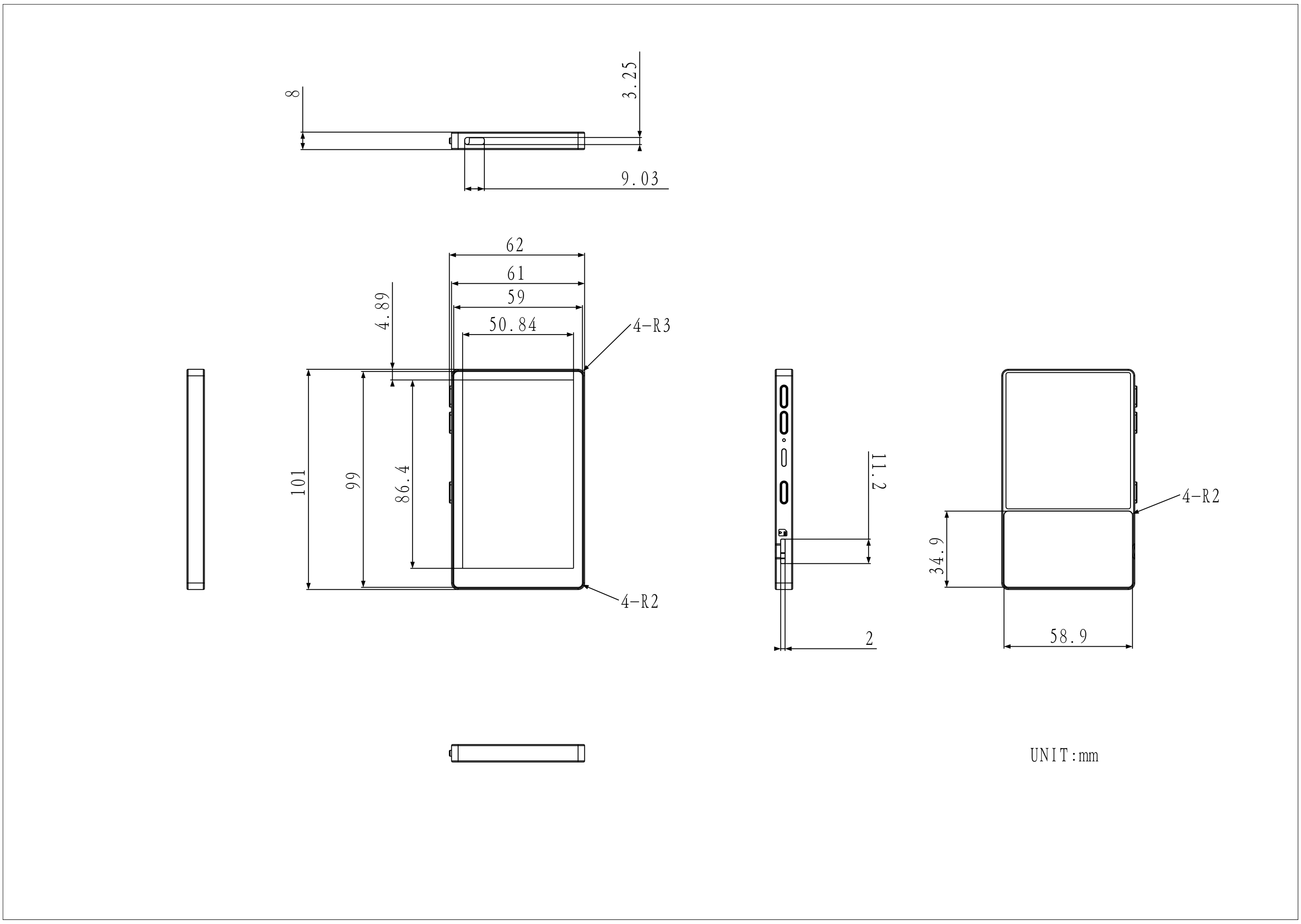
ESP32-S3R8	G38_SPI1_MOSI	G40_SPI1_MISO	G39_SPI1_CLK	G41	G21	G5
Stamp LoRa-1262	SPI_MOSI	SPI_MISO	SPI_CLK	SX_NSS	SX_BUSY	LORA_IRQ

M5IOE1	PYG10_PWM4	PYG2_ADC1
Stamp LoRa-1262	SX_NRST	SX_ANT_SW

M5PM1	G2
Stamp LoRa-1262	LoRa_EN

Model Size

- PaperMono Model Size PDF


8
3.25
9.03
62
61
59
50.84
4-R3
4.89
101
99
86.4
4-R2
11.2
2
34.9
4-R2
58.9
UNIT:mm

Datasheets

- ESP32-S3

- NFC ST25R3916
- SSD1677 Display Driver
- 3.97-inch Touchscreen
- RX8130CE Register Datasheet
- BMI270
- SX1262

Softwares

E-Paper Driver Notes

1. The e-paper driver waveforms for PaperMono in the M5GFX library are currently unstable. It is recommended to prioritize the e-paper manufacturer's OTP example below for refresh configuration to achieve better panel life and refresh stability.
2. Software recommendation: After approximately 10 partial fast refreshes, perform one full-screen refresh to clear the screen and prevent ghosting from accumulating.
3. After prolonged repeated refreshing, black deposited pixels may appear on the panel due to the physical characteristics of the ink particles. Leave the panel idle for a while, then perform one full-screen refresh to restore it.

Arduino

- PaperMono Arduino Quick Start
- PaperMono M5PM1 & M5IOE1 Power Management
- M5PM1 Arduino Library
- M5IOE1 Arduino Library
- M5Unified Arduino Library
- M5GFX Arduino Library

UiFlow2

- PaperMono UiFlow2 Quick Start

Protocol

- M5PM1 Power Management Chip
- M5IOE1 IO Expander Management Chip

ESP-IDF

- PaperMono Factory Firmware Source Code
- PaperMono OTP Example
- PaperMono Low-Power Test Example

PlatformIO

```
[env:m5stack-papermono]
platform = espressif32@6.12.0
board = esp32-s3-devkitm-1
framework = arduino
board_build.partitions = default_16MB.csv
board_upload.flash_size = 16MB
board_upload.maximum_size = 16777216
board_build.arduino.memory_type = qio_opi
build_flags =
    -DESP32S3
    -DBOARD_HAS_PSRAM
    -mfix-esp32-psram-cache-issue
    -DCORE_DEBUG_LEVEL=0
    -DARDUINO_USB_CDC_ON_BOOT=1
    -DARDUINO_USB_MODE=1
lib_deps =
    M5Unified = https://github.com/m5stack/M5Unified#develop
    M5PM1 = https://github.com/m5stack/M5PM1
    M5IOE1 = https://github.com/m5stack/M5IOE1
    M5Unit-NFC=https://github.com/m5stack/M5Unit-NFC
    RadioLib = https://github.com/jgromes/RadioLib
```

Easyloader

Easyloader	Download	Note
PaperMono User Demo	download	/

Other

- PaperMono Factory Reset Firmware
- PaperMono CrossPoint E-Reader

Video

Product Comparison

Product Compare	 **PaperMono**	 **PaperMono-Lite**
Main Controller	ESP32-S3R8	ESP32-S3R8
Screen	3.97" 4-level grayscale monochrome e-paper, 480x800	3.97" 4-level grayscale monochrome e-paper, 480x800
Touch	FT6336G	FT6336G
Frontlight	Integrated	Integrated
Wi-Fi	2.4 GHz	2.4 GHz
NFC	ST25R3916	❌
LoRa	Stamp LoRa-1262	❌
Expansion Storage	microSD	microSD
IMU	BMI270	BMI270
RTC	RX8130CE	RX8130CE
Battery	1150mAh	1150mAh
Case Color	Gray	White

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